Slot - C | Assessment Tool - 3 Class Test | Shalini Douglas

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1) * Image enhancement in the spatial Domain :-

Image enhancement is the process of improving the visual appearance of an image or making important features easier to analyze. In the spatial domain, enhancement is performed directly on the pixels of the image. $g(x,y) = T[f(x,y)]$

(a) Basic Gray-Level Transformation :-

These transformations modify the intensity values

- Image Negative :- • Converts dark pixels into bright pixels and vice versa
  - • useful for enhancing white or bright details in dark region.
- Log Transformation : • Expands dark intensity values and compress bright values.
  - • useful for images containing important details in dark areas.
- Power-Law Transformation :- • Adjust brightness and contrast.
  - • useful for correcting under exposed or overexposed images.
- Contrast stretching :- • Expands a narrow range of intensity values
  - • useful when the captured image has poor contrast.

(b) Histogram Processing :- A histogram represents the distribution of intensity values in an image

- • Histogram equalization : Redistributes intensity values
- • Histogram specification :- • Adjusts the histogram to a desired distribution.
  - • It is useful when an image is too dark, too bright or has low contrast.

(c) Spatial filtering:-

It is a process a pixel using its neighbouring pixels.

- Mean filter: • Replaces a pixel with the average of neighbouring pixels
  • Reduces random noise.
- Median filter: • Replaces a pixel with the median value of its neighbors.
  • Very effective against salt and pepper noise.
- Laplacian filter: • Highlights rapid intensity changes.
  • Used for sharpening and detecting fine details

2) Image Smoothing Vs Image Sharpening:

* Difference

Image Smoothing	Image Sharpening:
* Reduces noise and unwanted details.	* Enhances edges and fine details
* Makes the image smoother	* Makes the image appear clearer and more detailed.
Suppresses high frequency components	* Enhances high frequency components.
Uses low pass filter	* Uses high pass filter.
May blur edges	* Makes edges more prominent.

Spatial Smoothing filters:- Smoothing filters operate on a neighborhood around each pixel

- Mean filter:- The output pixel is calculated using the average of neighboring pixels.

$$g(x,y) = \frac{1}{mn} \sum f(s,t)$$

Median filter :- • The neighboring pixels values are sorted and the middle value is selected.

Eg:- 10, 12, 13, 255, 14

median = 13

• It is particularly effective for salt and pepper noise better than mean filter.

*** Spatial Sharpening filters** :- It emphasizes sudden intensity changes such as edges.

• **Laplacian filter** :- It detects rapid intensity changes

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

∴ the result can be combined with the original image to obtain sharper image.

• **High pass filter** :- High frequency components corresponding to edges and fine details are emphasized.

*** Applications** :-

• Smoothing should be applied when the image contains noise

• Sharpening should be applied when edges are weak.

3) *** Image enhancement in the Frequency Domain** :-

In frequency domain enhancement, an image is first converted from the spatial domain into the frequency domain using the fourier transform.

$$f(x,y) \longrightarrow F(u,v) \longrightarrow H(u,v)\,F(u,v) \longrightarrow G(u,v)$$

*** Low-Pass filters** :- It allows low frequency components to pass and suppress high frequency components

• It represents smooth regions, General Image information.
• Noise reduction, preprocessing before segmentation

* High-Pass filters :-

• High pass filters allow high frequency components
• Effects are enhances edges, Enhances fine details, Improves image sharpness, may also increase noise
• Applications are edge enhancement, feature extraction, text enhancement, defect detection.

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4) Image Segmentation :-

It is the process of dividing an image into meaningful regions or objects so that important features can be analyzed separately.

• Point Detection :- It identifies isolated pixels whose intensity differs significantly from their surroundings.

→ Applications :- Detecting small defects,
spots,
isolated bright pixels,
isolated dark pixels.

• Line Detection :- Special masks are used to detect lines in different directions.

→ Example :- Horizontal lines, Vertical lines, 45° lines.
→ Applications :- Detecting cracks, roads,
document lines.

• Edge Detection :- It identifies locations where intensity changes rapidly.

→ common :- Sobel,
Prewitt,
Roberts,
canny.

Thresholding :- Threshold T divides an image into two or more regions.

$$g(x,y) = \begin{cases} 1, & f(x,y) > T \\ 0, & f(x,y) < T \end{cases}$$

- It is very effective when the object and background have clearly different intensities.

- **Region-based Segmentation :-** pixels having similar properties are region growing, region splitting, Region merging.

* Edge Detection, Edge Linking and Boundary Detection :-

These techniques are important because they help identify the shape and boundaries of objects in an image.

* Edge Detection :-

- An edge is a location where there is a significant change in image intensity.
- It identifies sobel operator, Prewitt operator, roberts operator, canny edge detector.
- Detecting object boundaries and manufacturing defects.

* Edge lighting :-

- Edge detection may produce small disconnected edge segments.
- Edge linking connects these segments to form continuous boundaries.

- It considers :
  - pixel connectivity.
  - Direction.
  - Distance.

Eg:- ---- ---- (Before Linking)

---- -------- (After Linking)

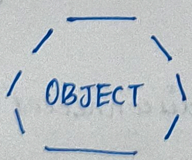
* Boundary Detection :-

- Boundary detection identifies the closed or outer boundary of an object.

- A process is :- original image
  ↓
  Edge Detection
  ↓
  Edge Linking
  ↓
  Boundary formation
  ↓
  Object Recognition.

- It provides
  - shape,
  - size
  - position
  - Orientation
  - Object Structure.

- Eg :-



boundary.